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Bayesian Generalized Hyperbolic Secant GLMs

PROJECT FOR THE BAYESIAN STATISTICS COURSE

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Abstract: We present a study of the Generalized Hyperbolic Secant (GHS) distribution, an overlooked member of the well-known Natural Exponential Family with Quadratic Variance Function (NEF-QVF) class. In this work, beyond a detailed theoretical analysis, we develop powerful and scalable Markov Chain Monte Carlo (MCMC) algorithms to sample from the posterior of a GHS Generalized Linear Model. We demonstrate that this model provides a meaningful alternative to the standard Gaussian linear model when dealing with heavy tails and heteroskedasticity. Our results, based on both simulated datasets and real-world applications, highlight the potential and the limitations of the GHS distribution in various statistical contexts.

Key-words: Generalized Hyperbolic Secant, Markov Chain Monte Carlo, MALA

Contents

1	Introduction	3
2	Theoretical Formulation	3
2.1	Hyperbolic Secant	4
2.2	Generalized Hyperbolic Secant	4
3	Hyperbolic Secant GLM	5
3.1	Metropolis-Hastings MCMC	6
3.2	MALA Algorithm	6
4	Extending the algorithms to the Generalized Hyperbolic Secant GLM	7
4.1	Metropolis-within-Gibbs algorithm	8
4.2	Adding a MALA step to improve mixing	8
5	Application to Real-World Data: Regressing Temperature Variation at New York Airports using atmospheric data	9
5.1	Heteroskedasticity	10
5.2	Fitting the Bayesian GLM	10
5.3	Evaluating model performance	11
5.4	Parameter inference	12
6	Model limitations	13
7	Conclusions	14
A	Natural Exponential Family with Quadratic Variance Function	16
B	Derivation of the Gamma formulation for the density of the GHS	16
C	Sampling directly from the GHS	16

1. Introduction

Bayesian Generalized Linear Models (GLMs) extend the standard Bayesian Gaussian linear model by allowing the response variable to follow distributions other than the normal. They provide a flexible framework for inference on regression parameters when Gaussian assumptions—such as homoskedasticity and normal errors—are violated. However, GLMs are defined only when the response distribution belongs to the exponential family. This restriction can be limiting in practice.

In particular, there is no widely established GLM for situations in which the response variable is continuous, heteroskedastic, and takes both positive and negative values—yet such settings are quite common. Examples include modeling ratios, log-errors, differences, and other variation measures. In these cases, the variance often increases approximately quadratically with the magnitude of the response.

A natural candidate for modeling this type of behavior is the Generalized Hyperbolic Secant (GHS) distribution. Introduced by Morris (1982), the GHS completes the class of natural exponential family distributions with quadratic variance function. Despite its attractive theoretical properties, the GHS has received limited attention in the statistical literature, mainly due to the complexity of its analytical expression and its relatively niche applications.

In this project, we develop Markov Chain Monte Carlo (MCMC) algorithms to perform parameter inference within a Bayesian GLM framework based on the Generalized Hyperbolic Secant distribution. We then assess the practical value of this model by applying it to a real dataset and evaluating its empirical performance. All of the necessary code was written using R programming language.

2. Theoretical Formulation

Many of the most commonly used distributions in statistical modeling belong to the *Exponential Dispersion Family* (EDF). A particularly important subclass is the *Natural Exponential Family* (NEF), characterized by densities expressed in canonical form with respect to a natural parameter. Within this class, special attention is given to distributions whose variance is a quadratic function of the mean, known as the *Natural Exponential Family with Quadratic Variance Function* (NEF–QVF).

A classical result by Morris (1982) shows that only six distributions belong to the NEF–QVF class: the Gaussian, Poisson, Binomial, Gamma, Negative Binomial, and the Generalized Hyperbolic Secant (GHS). These distributions are of particular interest because the quadratic relationship between mean and variance provides a structured yet flexible framework for modeling heteroskedastic data.

The first five distributions are well known and widely used in both classical and Bayesian generalized linear models. In contrast, the Generalized Hyperbolic Secant distribution has received comparatively little attention, despite sharing the same theoretical foundation. The GHS can be viewed as a generalization of the Hyperbolic Secant (HS) distribution through the introduction of an additional dispersion parameter, which increases modeling flexibility while preserving the quadratic variance structure.

An important structural feature of the NEF–QVF class is the support of its members. Among these distributions, the Gaussian is the only one defined on the entire real line. However, although it has full real support, it is symmetric and assumes constant variance in standard linear models. This implies that it does not naturally accommodate heteroskedasticity through its variance function, nor does it capture heavy tails or asymmetry without additional modelling adjustments.

In applications where the response variable exhibits skewness, heavy tails, or variance that increases with the magnitude of the mean, the Gaussian assumption may be inappropriate. A common practical remedy is to transform the response variable (for example, using logarithmic or Box–Cox transformations). While such approaches may reduce heteroskedasticity, they alter the interpretability of the model and do not directly address the underlying variance structure.

For these reasons, distributions such as the Hyperbolic Secant and its generalized version provide a theoretically grounded alternative within the NEF–QVF framework, allowing for quadratic variance behavior while maintaining full real support.

2.1. Hyperbolic Secant

A complete understanding of the HS and GHS distributions is incomplete without a general introduction to the NEF-QVF framework (found in appendix A).

In this framework, suppose that Y is distributed according to the Hyperbolic Secant: $Y | \theta \sim \mathcal{HS}(\theta)$. Its density function is given by

$$f(y; \theta) = \frac{\exp(\theta y + \log \cos(\theta))}{2 \cosh(\pi y/2)}, \quad (1)$$

where $y \in \mathbb{R}$ and $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Referring to formulation (33) of the exponential family distributions, we have:

$$b(\theta) = -\log(\cos \theta) \quad c(y, \phi) = c(y) = -\log(2 \cosh(\pi y/2)) \quad a(\phi) = 1 \quad (2)$$

Where a is the dispersion function, b is the cumulant function and c is the base measure of the exponential family. All notation is appropriately defined in appendix A.

This means the HS belongs to the NEF and its mean and variance functions are:

$$\mu(\theta) = \tan(\theta) \quad v(\mu) = b''(\arctan(\mu)) = 1 + \mu^2. \quad (3)$$

Hence, the HS has a quadratic variance function and thus belongs to the NEF-QVF class.

The HS distribution represents an interesting alternative to the Gaussian for modeling skewness and heavy-tailed phenomena, as can be seen in Fig. 1.

In particular, we want to use it as the response distribution of Bayesian GLM. However, as stated, the HS distribution only has one parameter, preventing a specific control of the level of dispersion in the tails. In order to build a GLM with this distribution, we have to introduce another parameter, so we have control over both the mean and the variance of the random variable. This generalization is provided by the GHS distribution.

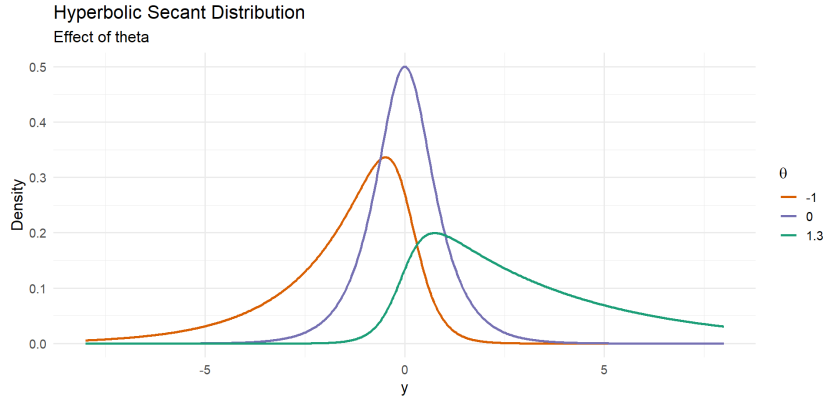


Figure 1: Hyperbolic Secant density function

2.2. Generalized Hyperbolic Secant

A more complete modelling of skewness and heteroskedasticity can be achieved by introducing a dispersion parameter $r > 0$ to the formulation of the HS distribution. This defines the Generalized Hyperbolic Secant ($Y|\theta, r \sim \mathcal{GHS}(\theta, r)$). In particular, the density function as introduced by Morris in 1982 is:

$$f_{r,\theta}(y; \theta, r) = \frac{2^{r-2} \Gamma(\frac{r}{2})^2}{\pi \Gamma(r)} \prod_{j=0}^{\infty} \left(1 + \frac{y^2}{(r+2j)^2} \right)^{-1} \exp(\theta y + r \log(\cos(\theta))) \quad (4)$$

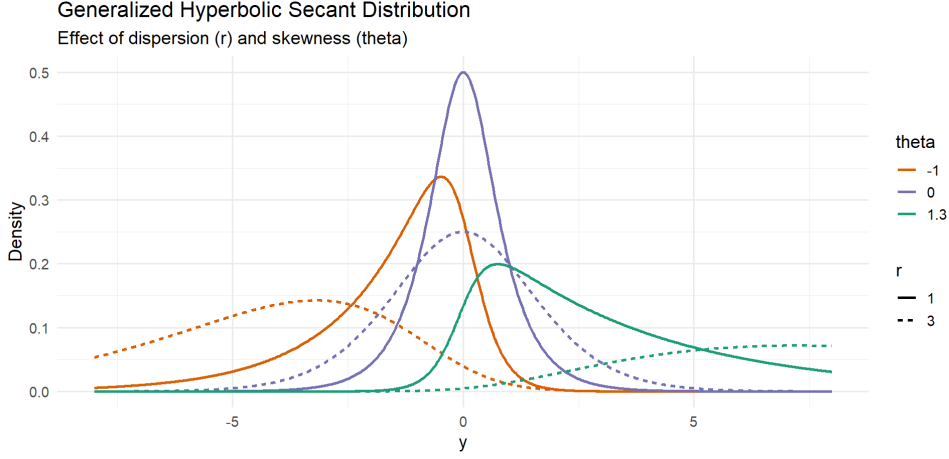


Figure 2: Generalize Hyperbolic Secant density

where $y \in \mathbb{R}$ and $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. The mean and variance function of Y in this case become:

$$\mu(\theta, r) = r \tan(\theta) \quad v(\mu) = b''(\arctan(\mu/r)) = r + \frac{\mu^2}{r}. \quad (5)$$

$$= r + r \tan(\theta)^2 \quad (6)$$

Notice that introducing r adds flexibility by allowing us to control both the mean and the variance. The HS was essentially a special case of the GHS with $r = 1$. As shown in Figure 2, the dispersion parameter r affects the location of the density and the amount of mass in the tails. The parameter θ still controls the skewness.

The main issue with the GHS is that its density function involves an infinite product. The lack of a closed-form expression makes it difficult to handle numerically. We could truncate the infinite product to approximate the value of the density, however it would be computationally expensive and therefore it would cause inefficiency in MCMC algorithms. Furthermore, it would make computations less exact. To overcome this issue, we found in the literature an alternative formulation that was not mentioned by Morris¹:

$$f_{r,\theta}(y; \theta, r) = \frac{2^{r-2}}{\pi \Gamma(r)} \left| \Gamma \left(\frac{r + jy}{2} \right) \right|^2 \exp(\theta y + r \log(\cos(\theta))) \quad (7)$$

with $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $r > 0$ and $j \in \mathbb{C}$ denoting the imaginary unit. A more detailed derivation of (7) is stated in Appendix B.

This formulation is actually going to be essential for our algorithm because approximating this value in R is much more efficient and accurate. To approximate evaluations of the complex gamma function we used the 'gammaz' function from the 'pracma' library in R.

3. Hyperbolic Secant GLM

After the in-depth theoretical study of both the HS and GHS distributions, we proceeded to implement an MCMC algorithm in R to sample from a Bayesian GLM with HS response. Although this model is not directly applicable to a real dataset due to the lack of a parameter controlling dispersion, it represents a useful preliminary step before introducing the generalized version.

The first step of rigorously defining a GLM is the link function $g: \mathbb{R} \rightarrow \mathbb{R}$, which relates the mean μ_i and the linear predictor η_i of an observation i . Namely, $g(\mu_i) = \eta_i = \mathbf{x}_i^t \beta$ where $\mathbf{x}_i$ are the covariates

¹We will denote with $\Gamma(z)$ the gamma function $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$. Instead for the gamma distribution we will use $\text{Gamma}(a, b)$.

for observation i . Since the mean of a Hyperbolic Secant random variable already has support on the real line (refer to mean function (3)), the most reasonable link function is the identity:

$$\mu_i = \eta_i \Rightarrow \tan(\theta_i) = \mathbf{x}_i^T \boldsymbol{\beta} \quad (8)$$

where the second equality follows from Section 2.1.

We did not choose the canonical link ($\theta(\mu_i) = \eta_i$) because the natural parameter of the HS and GHS is constrained to the set $(-\frac{\pi}{2}, \frac{\pi}{2})$, but obviously there is no guarantee that the linear predictor belongs to this set.

Next, we need to choose a prior for $\boldsymbol{\beta}$. For simplicity, we consider a multivariate Gaussian prior. In view of these considerations, the full Bayesian GLM is:

$$\begin{aligned} y_i | \theta_i &\stackrel{\text{ind}}{\sim} \mathcal{HS}(\theta_i) \\ \theta_i &= \arctan(\mathbf{x}_i^T \boldsymbol{\beta}) \\ \boldsymbol{\beta} &\sim \mathcal{N}_p(\mathbf{m}_0, \Sigma_0) \end{aligned} \quad (9)$$

We start by considering the simplest possible MCMC algorithm to sample from the posterior of $\boldsymbol{\beta}$.

3.1. Metropolis-Hastings MCMC

In the framework just presented, we write the likelihood of $\boldsymbol{\beta}$ (using the density (1)). For a sample of n independent observations, we get:

$$L(\boldsymbol{\beta} | \mathbf{y}, \mathbf{X}) = \prod_{i=1}^n \frac{\exp(y_i \arctan(\mathbf{x}_i^T \boldsymbol{\beta}) + \log \cos \arctan(\mathbf{x}_i^T \boldsymbol{\beta}))}{2 \cosh(\pi y_i / 2)} \quad (10)$$

By multiplying the Gaussian prior we get the posterior. Ignoring constants independent of $\boldsymbol{\beta}$ (such as the $2 \cosh(\pi y_i / 2)$ term), and using the identity $\cos(\arctan(u)) = (1 + u^2)^{-1/2}$, the log-posterior $\ell(\boldsymbol{\beta})$ is:

$$\ell(\boldsymbol{\beta}) \propto -\frac{1}{2}(\boldsymbol{\beta} - \mathbf{m}_0)^T \Sigma_0^{-1}(\boldsymbol{\beta} - \mathbf{m}_0) + \sum_{i=1}^n \left[y_i \arctan(\mathbf{x}_i^T \boldsymbol{\beta}) - \frac{1}{2} \log(1 + (\mathbf{x}_i^T \boldsymbol{\beta})^2) \right] \quad (11)$$

In order to sample from this posterior we use a Metropolis-Hastings approach. We use a symmetric Random Walk with proposal $\boldsymbol{\beta}^* | \boldsymbol{\beta}^{(k)} \sim \mathcal{N}(\boldsymbol{\beta}^{(k)}, \Sigma_{\text{rw}})$. The acceptance probability α is:

$$\alpha = \min \left(1, \frac{\exp\{\ell(\boldsymbol{\beta}^*)\}}{\exp\{\ell(\boldsymbol{\beta}^{(k)})\}} \right)$$

This works but there are many optimizations to be made.

3.2. MALA Algorithm

Unfortunately the random-walk Metropolis Hastings algorithm becomes very inefficient as the dimension of $\boldsymbol{\beta}$ (p) grows because the space to search becomes increasingly large.

A possible optimization is the Metropolis-Adjusted Langevin Algorithm (MALA) which uses the gradient of the log-likelihood to target regions of $\mathbb{R}^p$ with higher probability.

The proposal becomes:

$$\boldsymbol{\beta}^* | \boldsymbol{\beta}^{(k)} \sim \mathcal{N}_p \left(\boldsymbol{\beta}^{(k)} + \frac{\epsilon^2}{2} \nabla \ell(\boldsymbol{\beta}^{(k)}), \epsilon^2 \mathbf{I}_p \right) \quad (12)$$

In the specific case of the HS distribution we have:

$$\begin{aligned}\nabla \ell(\beta) &= \nabla \log \pi(\beta) + \sum_{i=1}^n [y_i - \tan(\theta_i)] \cdot \frac{\partial \theta_i}{\partial \beta} \\ \text{with : } \quad \frac{\partial \theta_i}{\partial \beta} &= \frac{\mathbf{x}_i}{1 + (\mathbf{x}_i^T \beta)^2}\end{aligned}\tag{13}$$

Since we are no longer working with a random walk MCMC; the proposal distribution $q(\beta_{\mathbf{a}}|\beta_{\mathbf{b}})$ (formula (13)) is no longer symmetric. So we adjust the acceptance rate α as follows:

$$\alpha = \frac{\exp\{\ell(\beta^*)\} q(\beta^{(k)} | \beta^*)}{\exp\{\ell(\beta^{(k)})\} q(\beta^* | \beta^{(k)})}\tag{14}$$

The most challenging part of implementing this procedure is that the results are extremely sensitive to the hyperparameter ϵ (formula (12)), which is a well-known issue of MALA. The most commonly adopted solution, which we also used, is to adaptively tune ϵ during the burn-in iterations. The parameter can be adjusted to target an ideal acceptance rate: increasing ϵ decreases the acceptance rate, whereas decreasing ϵ increases it. The literature (specifically [6]) suggests that the optimal acceptance rate for MALA in high-dimensional settings for β is approximately 0.574.

With this adaptive tuning, the algorithm performs very well. A comparison of results between the random-walk and MALA procedures is presented later.

4. Extending the algorithms to the Generalized Hyperbolic Secant GLM

As mentioned in Sec. 3, in order to obtain a usable GLM, the algorithms described previously are not sufficient. We now generalize the framework developed in Sec. 3 to a final GLM with GHS response. We still use an identity link function. With reference to the GHS mean function (5), this implies that the linear predictor is defined as

$$\mu_i = r \tan(\theta_i) = \eta_i = \mathbf{x}_i^T \beta.\tag{15}$$

Extending model (9), the full Bayesian GLM becomes

$$\begin{aligned}y_i | \theta_i, r &\stackrel{\text{ind}}{\sim} \mathcal{GHS}(\theta_i, r) \\ \theta_i &= \arctan\left(\frac{\mathbf{x}_i^T \beta}{r}\right) \\ \beta &\sim \mathcal{N}_p(\mathbf{m}_0, \Sigma_0) \\ r &\sim \text{Gamma}(a, b)\end{aligned}\tag{16}$$

It is important to understand how the model changes by introducing the parameter r . Although r is often referred to as a dispersion parameter, in the GHS GLM it plays a broader role. While the conditional mean remains $E[Y_i | X_i] = \mathbf{x}_i^T \beta$, the parameter r affects how the linear predictor is mapped into the natural parameter, and therefore influences the overall shape of the likelihood. In particular, r controls the strength of the quadratic mean–variance relationship: smaller values of r imply stronger heteroskedasticity and heavier tails, whereas larger values reduce the dependence of the variance on the magnitude of the mean. In the HS, without the parameter r , this relationship is fixed, which strongly limits the flexibility of the model.

We implemented this Bayesian model using two different MCMC approaches, described below. For the following algorithms, we reference the GHS density formulation using the complex Gamma function (7).

4.1. Metropolis-within-Gibbs algorithm

Evidently, we now have two parameters to estimate simultaneously: β (a multivariate parameter in $\mathbb{R}^p$) and r (a univariate parameter in $(0, \infty)$).

The most straightforward approach is to alternatively sample β and r using a Metropolis-within-Gibbs algorithm.

In each iteration $k + 1$, we update the parameter r conditionally on the most recent values of the betas, $\beta^{(k+1)}$.

As before, we propose β^* using

$$\beta^* | \beta^{(k)} \sim \mathcal{N}_p \left(\beta^{(k)}, \Sigma_{rw} \right). \quad (17)$$

Where $\Sigma_{rw} \in \mathbb{R}^{p \times p}$ is the prior covariance matrix. We propose a new shape parameter $r^* > 0$ from a Gamma distribution parameterized such that its mean is the current value $r^{(k)}$:

$$r^* | r^{(k)} \sim \text{Gamma} \left(\frac{(r^{(k)})^2}{\delta^2}, \frac{r^{(k)}}{\delta^2} \right), \quad (18)$$

where δ^2 is a tuning parameter controlling the variance of the proposal.

Because the Gamma proposal is asymmetric, $q(r^* | r^{(k)}) \neq q(r^{(k)} | r^*)$. The acceptance probability α_r is

$$\alpha_r = \min \left(1, \frac{\pi(r^* | y, \beta^{(k+1)})}{\pi(r^{(k)} | y, \beta^{(k+1)})} \times \frac{q(r^{(k)} | r^*)}{q(r^* | r^{(k)})} \right). \quad (19)$$

The proposal ratio $\frac{q(r^{(k)} | r^*)}{q(r^* | r^{(k)})}$ is calculated using the Gamma PDF $f_G(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$:

$$\frac{q(r^{(k)} | r^*)}{q(r^* | r^{(k)})} = \frac{f_G \left(r^{(k)}; \frac{(r^*)^2}{\delta^2}, \frac{r^*}{\delta^2} \right)}{f_G \left(r^*; \frac{(r^{(k)})^2}{\delta^2}, \frac{r^{(k)}}{\delta^2} \right)}. \quad (20)$$

Note that an alternative approach would be to sample $\log r \in (-\infty, \infty)$ using a Gaussian random walk. This would slightly simplify the acceptance probability formula.

The main issue with this procedure is the random-walk step on β . When the dimension p increases, the size of the space to search grows exponentially, and the mixing of the MCMC deteriorates dramatically. To make the model more scalable, further improvements are necessary.

4.2. Adding a MALA step to improve mixing

As we showed with the HS GLM in Sec.3.2 the MALA algorithm can be a significant improvement. Therefore, the algorithm we constructed is a kind of Gibbs sampler alternating between a MALA sampler for β (as explained in Sec. 3.2) and a Metropolis-Hastings step on r (as explained in Sec. 4.1). With the introduction of the parameter r , the formulas from Sec.3.2 need to be adapted. We now use the GHS density, so the gradient of the log-likelihood of β (formula (13)) has a slightly different expression. Using the new likelihood from expression (7), we obtain:

$$\nabla_\beta \ell(\beta | r, y, X) = -\Sigma_0^{-1}(\beta - m_0) + \sum_{i=1}^n (y_i - x_i^\top \beta) \frac{x_i}{r \left[1 + \left(\frac{x_i^\top \beta}{r} \right)^2 \right]} \quad (21)$$

The Metropolis-Hastings step on r requires no modification, since it is a scalar. Any changes would add complexity with little performance gain.

To illustrate the improvement in mixing for β due to MALA, we compared the two MCMC algorithms on high-dimensional simulated data ($p = 10$). We simulated a dataset of size $n = 100$ according

to the GHS. Arbitrary values for $\beta \in \mathbb{R}^{10}$ and $r \in (0, \infty)$ were fixed, and the covariates $\mathbf{x}_i \in \mathbb{R}^{10}$ were generated randomly for all $i = 1, \dots, n$. The response was then sampled according to the GHS distribution:

$$\begin{aligned} y_i | \theta_i, r &\sim \mathcal{GHS}(\theta_i, r) \\ \theta_i &= \arctan\left(\frac{\mathbf{x}_i^T \beta}{r}\right) \end{aligned} \quad (22)$$

$\forall i = 1, \dots, n$. Moreover, the sampling from the GHS was performed using a rejection algorithm (explained in detail in appendix C).

We ran both the random-walk MCMC algorithm and the version with MALA, using the same priors and hyperparameters. To highlight the improvement in mixing, the autocorrelation plots for selected coefficients are shown in Fig. 3.

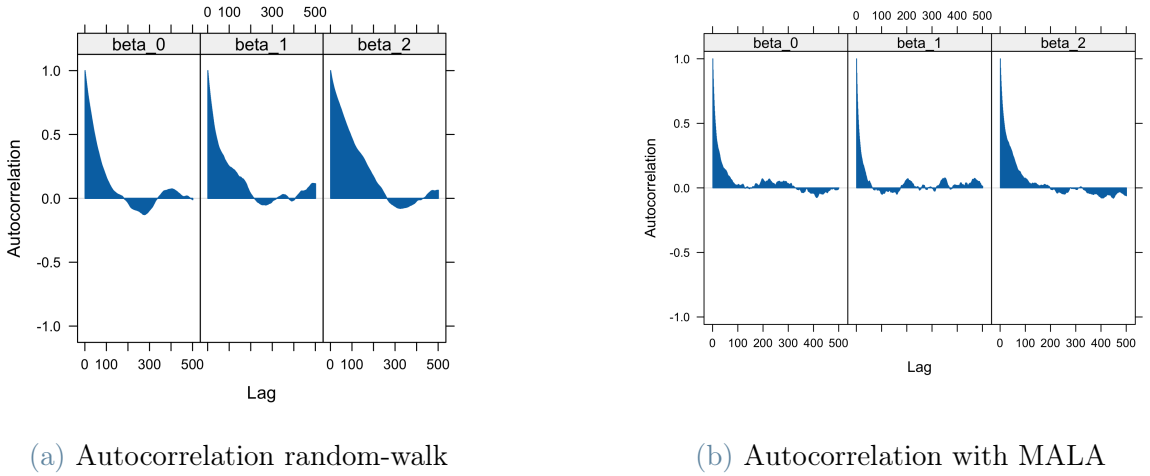


Figure 3: Autocorrelation plots

5. Application to Real-World Data: Regressing Temperature Variation at New York Airports using atmospheric data

Fitting our model on simulated data is not sufficient to demonstrate the effectiveness of the algorithm; we need to validate it on a real-world dataset.

As previously mentioned, the GHS distribution is well suited for modeling heteroskedasticity in empirical data in many contexts. In the case of GLMs there is no alternative for modelling a heavy-tailed response taking both positive and negative real values. To assess the models' applicability to real data and its convenience with respect to the Gaussian distribution we decided to test it on a dataset containing hourly meteorological observations recorded at New York City airports in 2013 taken from the `nycflights13` package in R. This dataset also provides a challenge to the MCMC efficiency and scalability, as it consists of several regressors and a large sample size ($n = 26155$).

For the present study, the regressors are atmospheric variables, namely temperature (`temp`), dew point (`dewp`), relative humidity (`humid`), wind speed (`wind speed`), precipitation (`precip`), and atmospheric pressure (`pressure`).

The absolute temperature level is difficult to be inferred and may depend on many other factors; therefore, we use the hourly temperature change instead. The response variable is defined as the difference in the `temp` variable: $\Delta temp_t = temp_t - temp_{t-1}$. As explanatory variables, we include the

one-period lagged values of all selected meteorological variables². This specification yields a regression framework in which temperature variations are modeled as a function of past atmospheric conditions:

$$\Delta temp \sim temp + dewp + humid + wind_speed + precip + pressure \quad (23)$$

5.1. Heteroskedasticity

The traditional approach would be to fit a standard Gaussian linear model or a Bayesian linear model. The plots below show diagnostic checks for OLS regression. Specifically, plot 4a illustrates heteroskedasticity, with residual variance increasing for both large and small values of the fitted response. For a quantitative assessment, we performed a studentized Breusch-Pagan test for heteroskedasticity and obtained $p\text{-value} < 2.2 \times 10^{-16}$, conclusively indicating the presence of heteroskedasticity.

Plot 4b shows a Gaussian Q-Q plot, confirming that the response variable exhibits heavy tails. The Gaussian assumptions of linear models are clearly violated, and therefore the corresponding inference on β would be unreliable.

Considering this preliminary analysis, we conclude that attempting a Bayesian GHS GLM is appropriate.

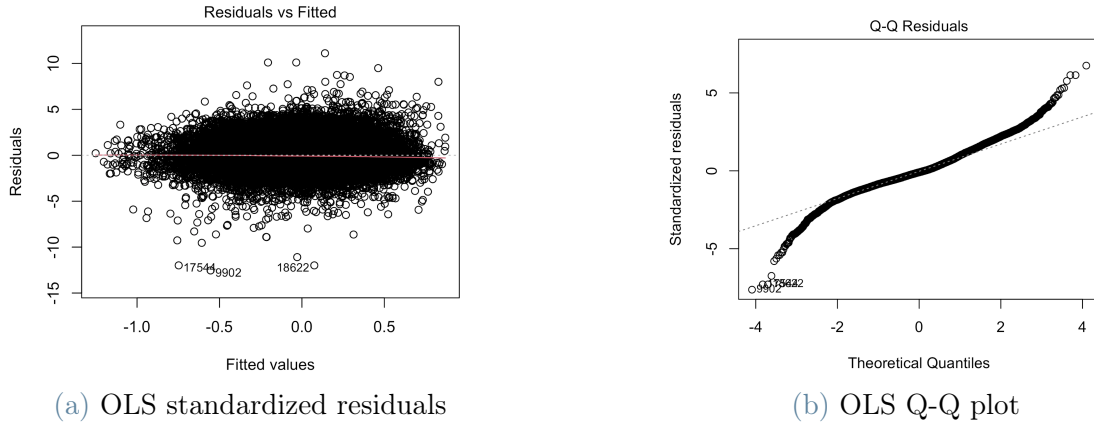


Figure 4: OLS plots

5.2. Fitting the Bayesian GLM

In reference to the full Bayesian GHS GLM formulation (16), the complete model we fitted for this dataset is:

$$y_i | \theta_i, r \stackrel{\text{ind}}{\sim} \mathcal{GHS}(\theta_i, r) \quad (24)$$

$$\theta_i = \arctan \frac{\mathbf{x}_i^T \boldsymbol{\beta}}{r} \quad (25)$$

$$\boldsymbol{\beta} \sim \mathcal{N}_p(\mathbf{0}, 5\hat{S}) \quad (26)$$

$$r \sim \Gamma(\epsilon, \epsilon) \quad (27)$$

where $\hat{S}$ is the OLS covariance matrix estimate which we chose to multiply by 5 to decrease prior information on $\boldsymbol{\beta}$. We chose $\epsilon = 0.1$ to ensure little prior information on r but maintain a prior mean of 1.

Since p (the dimension of $\boldsymbol{\beta}$) is 7 the random-walk Metropolis-Within-Gibbs MCMC algorithm expectedly performed quite poorly, there was insufficient mixing. We favoured the Metropolis-Within-Gibbs

²Observations with missing values induced by differencing and lagging are removed, and two additional anomalous rows are excluded to ensure numerical stability.

with a MALA step, which provided good results. The plots in Figure 5 show the traceplots for one of the coefficients β_i and for r using only 10,000 MCMC iterations with a burn-in of 1,000 iterations. The quality of the traceplot looks decent even for relatively few iterations. The corresponding effective sample sizes were $\approx 3,108$ and $\approx 2,543$ which is satisfactory. The quality of the MCMC sampling is of similar quality for all 7 betas.

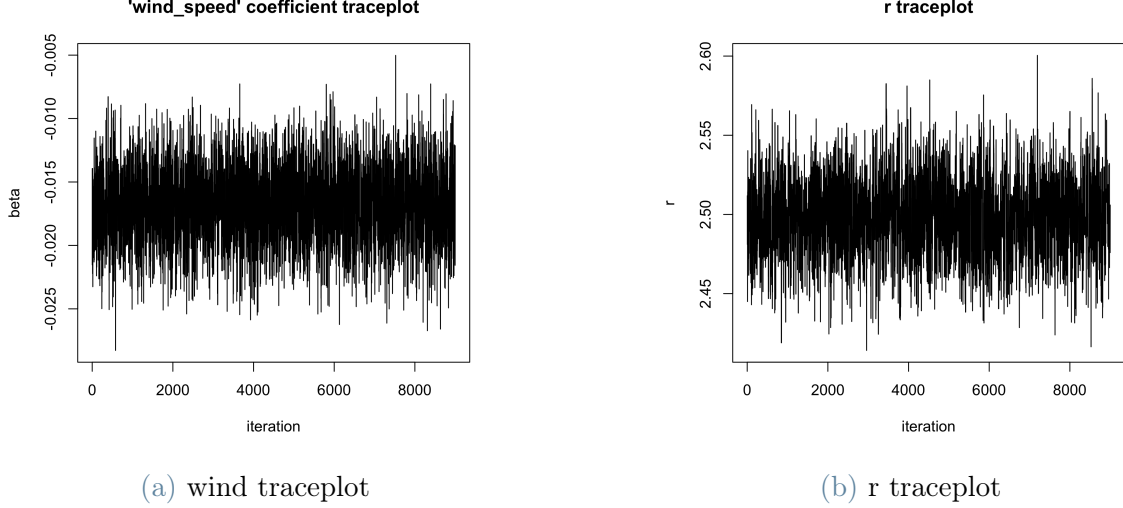


Figure 5: traceplots

5.3. Evaluating model performance

To assess the validity of our model, we compared the results of the GHS GLM with those of a Bayesian linear model using the same prior for β .

Firstly for both models we computed the Widely Applicable Information Criterion (WAIC) from posterior draws. WAIC is computed from the *pointwise* log-likelihood evaluated across posterior samples and therefore requires, for each model, a matrix

$$[\log p(y_i | \theta^{(s)})]_{s=1, \dots, S; i=1, \dots, n},$$

where S is the number of posterior draws and n is the number of observations. For each posterior draw $(\beta^{(s)}, r^{(s)})$, we compute the linear predictor and induced natural parameter

$$\eta_i^{(s)} = x_i^\top \beta^{(s)}, \quad \theta_i^{(s)} = \arctan\left(\frac{\eta_i^{(s)}}{r^{(s)}}\right), \quad i = 1, \dots, n.$$

Using the gamma-function representation of the GHS density (Eq. (7)), the pointwise log-likelihood is

$$\log p(y_i | \theta_i^{(s)}, r^{(s)}) = \log f_{\text{GHS}}(y_i, \theta_i^{(s)}, r^{(s)}) \quad (28)$$

For the Bayesian linear model with posterior draws $(\beta^{(s)}, \sigma^{2(s)})$, the pointwise log-likelihood was computed as follows:

$$\log p(y_i | \beta^{(s)}, \sigma^{2(s)}) = \log \mathcal{N}(y_i; x_i^\top \beta^{(s)}, \sigma^{2(s)}), \quad i = 1, \dots, n,$$

Now given the pointwise log-likelihood matrix for a model, we define for each observation i :

$$\text{lpd}_i = \log \left(\frac{1}{S} \sum_{s=1}^S \exp(\log p(y_i | \theta^{(s)})) \right), \quad (29)$$

$$\text{pWAIC}_{i, \text{model}} = \text{Var}_{s=1, \dots, S}(\log p(y_i | \theta^{(s)})). \quad (30)$$

The WAIC estimate and its effective number of parameters are then

$$WAIC = -2 \sum_{i=1}^n (\text{lpd}_i - p_{WAIC,i}), \quad p_{WAIC} = \sum_{i=1}^n p_{WAIC,i}. \quad (31)$$

We got the following result:

$$\Delta WAIC = WAIC_{GHS} - WAIC_N = -741.79$$

which indicates that introducing the GHS provides a substantial improvement over the default Gaussian assumptions. This suggests that the GHS distribution offers a good approximation of the true heavy-tailed distribution of the response, which is important to ensure the reliability of the inferential conclusions provided by the model.

To further verify that the response distribution aligns with our assumptions, we computed the posterior predictive residuals. Using the variance function of the GHS, these residuals take the form:

$$\epsilon_i = \frac{y_i - \hat{\mu}_i}{\hat{\sigma}_i} = \frac{y_i - \hat{r} \tan \hat{\theta}_i}{\sqrt{\hat{r}(1 + \tan^2 \hat{\theta}_i)}}, \quad \forall i = 1, \dots, n. \quad (32)$$

Where $\hat{\mu}_i, \hat{\sigma}_i, \hat{r}, \hat{\theta}_i$ are the respective posterior estimates of $\mu_i, \sigma_i, r, \theta_i$.

Plot 6a shows that the model's mean and variance assumptions are essentially satisfied. There is no evident correlation between the fitted responses and the residuals.

Plot 6b compares the empirical quantiles of the residuals with the theoretical quantiles of the GHS distribution. Although the tails do not match perfectly, the GHS appears to be a satisfactory approximation of the true response distribution, and it is certainly much closer than the Gaussian. Therefore, we can conclude a posteriori that the model assumptions hold, and we can consider the parameter inference from the MCMC to be meaningful.

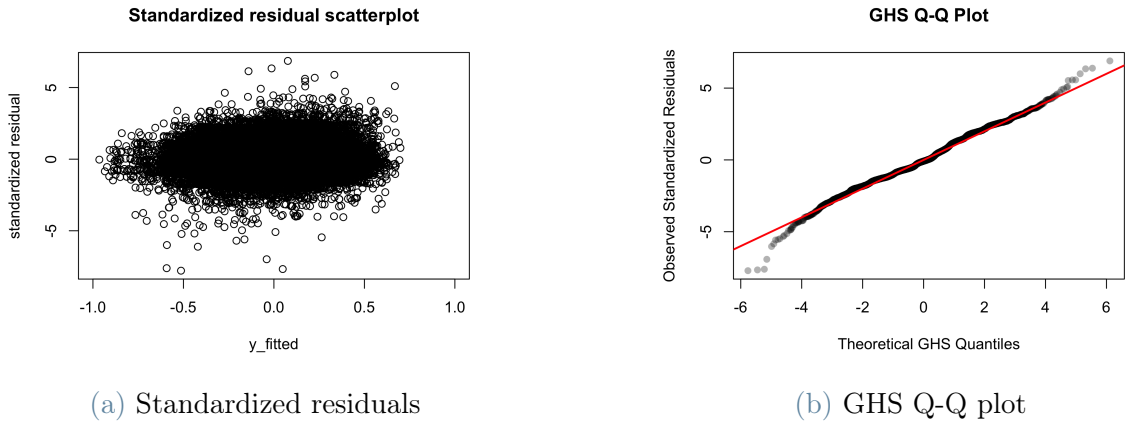


Figure 6: Distribution of residuals

5.4. Parameter inference

Below are reported the results of inference on each parameter β_i after running the MCMC simulation for 10 000 iteration with a burn-in of 1 000 iterations. Plot 7a represents the 95% posterior credible intervals for each coefficient. They were calculated by computing the 2.5% and 97.5% empirical quantiles of the MCMC posterior draws of β . The plot also reports the posterior mean of each parameter. Table 7b has the corresponding Bayes factors for the following test (to check meaningfulness of each covariate):

$$\begin{cases} H_0 : \beta_i = 0 \\ H_1 : \beta_i \neq 0 \end{cases} \quad \forall i = 1, \dots, 7$$

Since the null hypothesis is a point restriction, the Bayes factor in favor of H_0 can be written as

$$BF_{01} = \frac{\pi(\beta_i = 0 | y)}{\pi(\beta_i \neq 0)},$$

that is, the ratio between the posterior and prior marginal densities evaluated at zero. And $BF_{10} = 1/BF_{01}$, so that values greater than one indicate evidence against H_0 and in favor of the inclusion of the corresponding covariate.

In practice, the posterior density at zero was approximated using a kernel density estimate ('density' function in R) computed from the MCMC draws of β_i .

Therefore we can conclude that all covariates are meaningful with great certainty, except for **precip**. We could remove this regressor or use a different prior for variable selection however it is not relevant to pursue further analyses for the goal of this project. We merely wanted to prove that traditional inference metrics can be computed for this model as well.

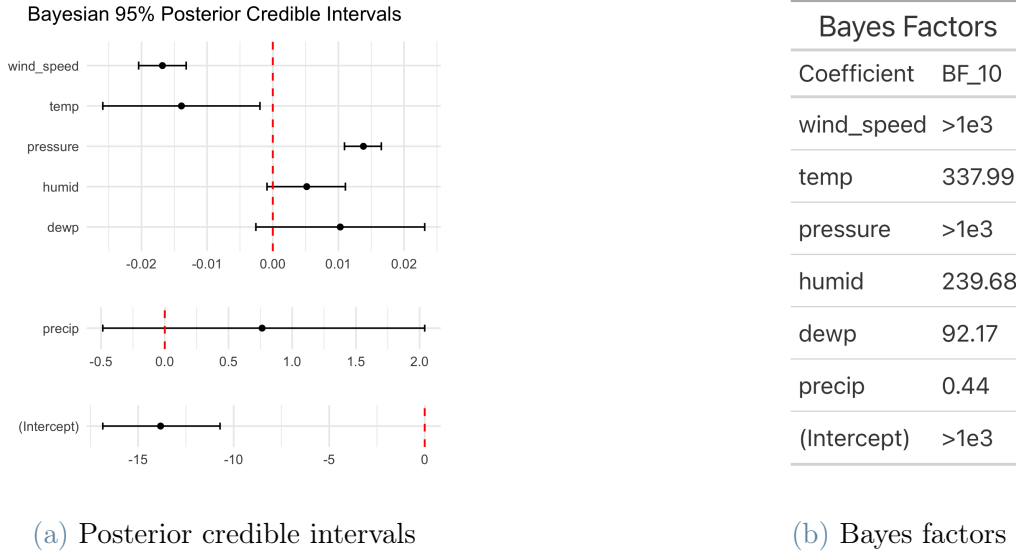


Figure 7: Beta inference

It is also interesting to study the posterior distribution of r . The posterior mean is ≈ 2.498 . The 95% posterior credible interval is (2.448, 2.549). This means it is very confidently greater than 1 (which is the value of r coinciding with the simple HS). If we had not introduced the GHS into the model we would not have been able to apply the model to this dataset.

Also this value of r is somewhat large which tells us that variance grows very rapidly as the mean of the response increases.

6. Model limitations

It is important to clarify that the GHS models a very specific kind of heteroskedasticity. The variance has to increase quadratically with the mean according to its specific variance function. This means that the Bayesian GHS GLM will not always outperform the corresponding Gaussian linear model in all instances of heavy-tailed responses. In fact we have seen for ourselves that the Gaussian framework is vastly more robust to hypothesis violations. If the true response distribution is not actually similar enough to the GHS, the Gaussian model always outperforms our GLM.

In fact, at first, we tried analyzing a different dataset. The goal was to model errors in the prediction of election outcomes across different polling agencies. Even though the variance structure seemed suitable, it was not similar enough to the GHS and our model performed poorly. For this reason we omitted the results of this regression from the project.

The Bayesian GHS GLM should be considered a tool that can, in some regressions, be the most accurate. However even when linear model assumptions are violated often the Bayesian Gaussian linear model remains better.

7. Conclusions

In this work we developed and implemented a Bayesian Generalized Linear Model based on the Generalized Hyperbolic Secant (GHS) distribution. The main objective was twofold: first, to construct efficient MCMC algorithms capable of sampling from the posterior of a GHS GLM in moderate-to-high dimension; second, to assess whether this model provides a meaningful alternative to the standard Gaussian linear model in the presence of heavy tails and heteroskedasticity.

From a methodological perspective, the main contribution lies in the implementation of scalable MCMC algorithms. A simple random-walk Metropolis–Hastings scheme was shown to mix poorly as the dimension of β increases. The introduction of a MALA step for β , combined with a Metropolis update for r , substantially improved mixing and effective sample size, even with relatively few iterations. The algorithm proved computationally feasible on a dataset with more than 20,000 observations, demonstrating that the model is practically implementable despite the non-standard likelihood.

From a modelling perspective, the GHS GLM provides a structured way to capture quadratic mean–variance relationships and heavy-tailed behavior while preserving support on the entire real line. Unlike the Gaussian model, where variance is independent of the mean unless explicitly modeled, the GHS distribution embeds heteroskedasticity directly into its variance function. This makes it particularly suitable in settings where variability naturally increases with the magnitude of the mean.

In the real world application we tested, the GHS GLM substantially outperformed the Gaussian model in terms of WAIC. Posterior predictive diagnostics indicated that the quadratic variance structure was consistent with the observed data, and the residual analysis suggested that the model assumptions were reasonably satisfied. The results support the idea that rapid and extreme temperature changes are intrinsically more volatile than stable conditions, a feature that the GHS model is structurally designed to capture.

At the same time, the model’s performance on alternative data (such as election prediction errors) revealed an important limitation: the GHS is not a generic heavy-tailed replacement for the Gaussian model. Its improvement depends critically on whether the data exhibit a variance structure compatible with the specific quadratic form implied by the distribution. When this structural assumption is not approximately satisfied, the Gaussian model may remain more robust and provide better overall performance.

Therefore, the Bayesian GHS GLM should be viewed not as a universal alternative to Gaussian regression, but as a specialized tool suited to contexts where heavy tails and variance–mean coupling are substantively justified. When such structure is present, the model can deliver substantial predictive gains and coherent inference. When it is absent, its additional complexity may not be warranted.

Overall, this project demonstrates that the GHS distribution, despite its analytical complexity and relative rarity in applied work, can be successfully integrated into a Bayesian GLM framework and estimated efficiently via modern MCMC techniques. The results highlight both the potential and the boundaries of this modeling approach.

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A. Natural Exponential Family with Quadratic Variance Function

The density function of a random variable $Y : \Theta \rightarrow \mathcal{Y}$ belonging to the exponential dispersion family can be written as

$$f(y_i; \theta, \phi) = \exp \left\{ \frac{\theta y_i - b(\theta)}{a_i(\phi)} + c(y_i, \phi) \right\}, \quad (33)$$

where $y_i \in \mathcal{Y} \subseteq \mathcal{R}$ is a realization of Y , $\theta \in \Theta \subseteq \mathcal{R}$, and $a_i(\phi) = \phi/\omega_i$, with ω_i known positive weights. The parameter θ is called the natural parameter, while $\phi > 0$ is the dispersion parameter. Moreover, we say that Y belongs to the natural exponential family (NEF) if $a_i(\phi) = 1$ and $c(y_i, \phi)$ does not depend on ϕ . In this case, the density can be written as

$$f(y_i; \theta) = \exp \{ \theta y_i - b(\theta) + c(y_i) \}. \quad (34)$$

The main properties of the EDF follow from the expressions of the mean and variance, which can be derived from the functions $a(\cdot)$ and $b(\cdot)$. In particular,

$$E[Y_i] = b'(\theta_i), \quad \text{Var}(Y_i) = a_i(\phi) b''(\theta_i). \quad (35)$$

Furthermore, the variance of a distribution belonging to the EDF can be expressed as a function of the mean. Defining $\mu(\theta) := E[Y_i] = b'(\theta)$ and letting $\theta(\mu_i) = (b')^{-1}(\mu_i)$, we can rewrite (35) as

$$\text{Var}(Y) = a(\phi) b''(\theta) = a(\phi) b''(\theta(\mu)) = a(\phi) v(\mu), \quad (36)$$

where $v(\mu) := b''(\theta(\mu))$ is called the variance function. Distributions with QVF are characterized by a variance function that has a quadratic relationship with the mean. This property plays a central role in our study.

B. Derivation of the Gamma formulation for the density of the GHS

Here we derive formula (7) starting from (4) given in Morris' paper. We use the following identity:

$$\prod_{j=0}^{\infty} \left(1 + \frac{y_i^2}{(r+2j)^2} \right)^{-1} = \frac{\left| \Gamma\left(\frac{r+jy_i}{2}\right) \right|^2}{\Gamma\left(\frac{r}{2}\right)^2} \quad (37)$$

Applying this identity to (4), including all normalization constants, we obtain:

$$f_{r,\theta}(y_i; \theta, r) = \frac{2^{r-2} \Gamma\left(\frac{r}{2}\right)^2}{\pi \Gamma(r)} \left[\frac{\left| \Gamma\left(\frac{r+jy_i}{2}\right) \right|^2}{\Gamma\left(\frac{r}{2}\right)^2} \right] \exp(\theta y_i + r \log(\cos(\theta))) \quad (38)$$

This gives exactly the formulation (7) stated previously. Identity (37) has been taken from [5], therefore we refer to it for further information.

C. Sampling directly from the GHS

Suppose we want to sample from the target GHS density $\pi(x)$ (formula (7)). The simplest way to do this is by using rejection sampling, which requires choosing a proposal density $m(x)$. We use a

Uniform distribution $m \sim \mathcal{U}(-M, M)$ since the GHS distribution has heavier tails than other bell-shaped distributions such as the Gaussian. In this way, we approximate the true distribution by simulating from a truncated version of the GHS on $[-M, M]$. Then, through numerical optimization, we find the smallest constant c such that:

$$\pi(x) \leq c \cdot m(x) \quad \forall x \in \mathcal{R} \iff \frac{\pi(x)}{c \cdot m(x)} \leq 1$$

The steps of the algorithm are:

1. **Simulate Candidate:** Generate a candidate point Z from the proposal distribution m .
2. **Compute Ratio:** Calculate the acceptance probability r :

$$r = \frac{\pi(Z)}{c \cdot m(Z)} \in (0, 1)$$

3. **Simulate Uniform:** Generate $U \sim \mathcal{U}(0, 1)$.
4. **Accept/Reject:**
 - If $U \leq r$, then we **accept** the candidate and set $\tilde{x} = Z$.
 - If $U > r$, then we **reject** Z and return to Step 1 to simulate a new candidate point independently.

This procedure produces values $\tilde{x}$ that are iid from an approximation of the GHS, but it is neither very efficient nor exact. However, running hundreds of different Metropolis–Hastings MCMCs would be even less efficient for generating many independent observations y_i with different parameter values θ_i .

In any case, this algorithm does not need to be particularly efficient because it is only used to simulate a dataset to compare the efficiency of different GLM algorithms in Sec. 4.2. If a more efficient method is needed, a good alternative would be the Ratio-of-Uniforms method. The algorithm works as follows:

1. **Compute** or approximate numerically the following quantities:

$$u_{max} := \sup_x \sqrt{\pi(x)}$$

$$v_{min} := \inf_x x \sqrt{\pi(x)}$$

$$v_{max} := \sup_x x \sqrt{\pi(x)}$$

2. **Sample** from uniform distributions:

$$U \sim \mathcal{U}(0, u_{max})$$

$$V \sim \mathcal{U}(v_{min}, v_{max})$$

$$X := \frac{V}{U}$$

3. **Accept/Reject:**
 - If $\ln(U) \leq \frac{1}{2} \ln \pi(X)$, then we **accept** the candidate and set $\tilde{x} = X$.
 - If $\ln(U) > \frac{1}{2} \ln \pi(X)$, then we **reject** X and return to Step 1 to simulate a new candidate point independently.

This procedure produces values $\tilde{x}$ that are iid from the exact GHS.